Couplet-Based Key to the Orders of Arachnida

1:
Opisthosoma (abdomen)
unsegmented or, if segmented, with spinnerets posteriorly on ventral side (Figures 5-5 and 5-8, ALS,PLS,PMS)

Option A: Opisthosoma petiolate (Figu...

2:
Opisthosoma petiolate (Figures 5-5 and 5-9 through 5-13)
Araneae
Order: Araneae

1':
Opisthosoma distinctly segmented, without spinnerets

Option A: Opisthosoma with a tail-lik...

2':
Opisthosoma not petiolate, but broadly joined to prosoma (Figures 5-15 through 5-22)
Acari
Order: Acari

3:
Opisthosoma with a tail-like prolongation that is either thick and terminating in a sting (Figure 5-3)
or slender and more or less whiplike (Figure 5-4A,B);
mostly tropical

Option A: Opisthosoma ending in a sti...

4:
Opisthosoma ending in a sting (Figure 5-3);
first pair of legs not greatly elongated;
second ventral segment of opisthosoma with a pair of comblike organs Scorpiones
Order: Scorpiones

3':
Opisthosoma without a tail-like prolongation or with a very short leaflike appendage

Option A: Pedipalps chelate (pincerli...

7:
Pedipalps chelate (pincerlike)
(Figure 5-23);
body more or less oval, flattened, usually less than 5 mm in length Pseudoscorpiones
Order: Pseudoscorpiones

4':
Opisthosoma not ending in a sting;
first pair of legs longer than other pairs (Figure 5-4A,B);
second ventral segment of opisthosoma without comblike organs

Option A: Pedipalps slender, similar ...

5:
Pedipalps slender, similar to legs (Figure 5-4A);
minute forms, 5 mm or less in length Palpigradi
Order: Palpigradi

7':
Pedipalps raptorial or leglike, but not chelate;
body not particularly flattened;
size variable

Option A: Chelicerae very large, usua...

8:
Chelicerae very large, usually about as long as prosoma, and extending forward, and body slightly narrowed in middle (Figure 5-4C);
color pale yellow to brownish;
length 20 to 30 mm;
mostly nocturnal desert forms occurring in western United States Solifugae
Order: Solifugae

5':
Pedipalps usually much stouter than any of the legs (Figure 5-4B);
mostly larger forms

Option A: With two median eyes on a t...

6:
With two median eyes on a tubercle anteriorly and a group of three eyes on each lateral margin;
tail long, filiform, and many-segmented;
pedipalps nearly straight or curved mesad, extending forward, and moving laterally (Figure 5-4B);
body blackish, more than 50 mm in length Uropygi
Order: Uropygi

8':
Without the preceding combination of characters

Option A: First pair of legs very lon...

9:
First pair of legs very long, with long tarsi;
opisthosoma constricted at base;
mainly tropical

6':
Eyes lacking;
tail short, one- to four-segmented;
pedipalps arching upward, forward, and downward and moving vertically;
body yellowish or brownish, less than 8 mm in length Schizomida
Order: Schizomida

Option A: Prosoma longer than wide, l...

9':
First legs similar to the others and (except in the Opiliones)
not usually long;
opisthosoma not particularly constricted at base

10:
Prosoma longer than wide, lateral margins nearly parallel, and with a transverse membranous suture in caudal third;
opisthosoma with a very short terminal appendage;
length 5-8 mm (see also couplet 6')
Schizomida
Order: Schizomida

Option A: Orange-red to brown arachni...

11:
Orange-red to brown arachnids, about 3 mm in length, with broad flap at anterior end of prosoma concealing chelicerae;
legs not unusually long, animal somewhat mitelike or ticklike in appearance;
recorded from Rio Grande Valley of Texas Ricinulei
Order: Ricinulei

10':
Prosoma wider than long, lateral margins rounded or arched, and without a transverse suture;
opisthosoma without a terminal appendage;
length variable, up to about 50 mm Amblypygi
Order: Amblypygi

11':
Size and color variable, but without broad flap at anterior end of prosoma covering chelicerae;
legs variable in length, often very long and slender (Figure 5-14);
widely distributed Opiliones
Order: Opiliones